



**សាកលវិទ្យាល័យភ្នំពេញអន្តរជាតិ**  
**PHNOM PENH INTERNATIONAL UNIVERSITY**

# Urban Cart E-Commerce Data Analysis Report

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# Table of Contents

|                                                                    |   |
|--------------------------------------------------------------------|---|
| 1. EXECUTIVE SUMMARY .....                                         | 1 |
| 1.1 Project Background .....                                       | 1 |
| 1.2 Strategic Questions .....                                      | 1 |
| 1.3 Analytical Approach.....                                       | 1 |
| 1.4 Main Analytical Evidence Available in the Supplied Report..... | 2 |
| 1.5 Management Value.....                                          | 2 |
| 2. INTRODUCTION AND BUSINESS CONTEXT .....                         | 3 |
| 2.1 Urban Cart .....                                               | 3 |
| 2.2 Why Business Context Matters .....                             | 3 |
| 2.3 Business Question 1: Customer Value and Loyalty .....          | 4 |
| 2.4 Business Question 2: Profitability and Demand.....             | 4 |
| 2.5 Business Question 3: Data Trustworthiness .....                | 4 |
| 3. DATA DESCRIPTION.....                                           | 5 |
| 3.1 Data Sources .....                                             | 5 |
| 3.2 Database Tables .....                                          | 5 |
| 3.3 External Customer Data.....                                    | 6 |
| 3.4 External Product Catalog.....                                  | 7 |
| 3.5 Data Lineage.....                                              | 7 |
| 3.6 Data Dictionary Considerations.....                            | 7 |
| 4. SQL EXTRACTION AND METHODOLOGY .....                            | 8 |
| 4.1 Revenue by Category.....                                       | 8 |
| 4.2 Top 20 Customers.....                                          | 8 |
| 4.3 Monthly Revenue Trend.....                                     | 8 |
| 4.4 Category Return Rate .....                                     | 8 |

|                                                 |    |
|-------------------------------------------------|----|
| 4.5 Consistent Customers .....                  | 9  |
| 4.6 Highly Rated Products .....                 | 9  |
| 4.7 Device Engagement .....                     | 9  |
| 4.8 Product Ranking Within Category .....       | 9  |
| 4.9 Payment Method Mix .....                    | 9  |
| 4.10 Additional Leadership Analysis .....       | 9  |
| 4.11 SQL Quality Controls .....                 | 10 |
| 5. DATA CLEANING AND INTEGRATION .....          | 10 |
| 5.1 Date Cleaning .....                         | 10 |
| 5.2 Before/After Evidence .....                 | 10 |
| 5.3 Duplicate Customers .....                   | 11 |
| 5.4 Normalization Principles .....              | 11 |
| 5.5 Missing Values .....                        | 11 |
| 5.6 Outliers .....                              | 12 |
| 5.7 Returns .....                               | 12 |
| 5.8 Product Reconciliation Workflow .....       | 12 |
| 5.9 Integration Validation .....                | 12 |
| 5.10 Processed Outputs .....                    | 13 |
| 5.11 Why Cleaning Can Change a Conclusion ..... | 13 |
| 5.12 What Was Not Fixed .....                   | 13 |
| 5.13 Reproducibility .....                      | 13 |
| 6. NUMPY ANALYTICAL METHODS .....               | 13 |
| 6.1 RFM Segmentation .....                      | 13 |
| 6.2 Interpreting RFM .....                      | 14 |
| 6.3 Customer-Product Interaction Matrix .....   | 14 |
| 6.4 Cosine Similarity .....                     | 14 |

|                                                              |    |
|--------------------------------------------------------------|----|
| 6.5 Recommendation Output.....                               | 14 |
| 6.6 Business Interpretation .....                            | 15 |
| 6.7 Normal-Equation Regression.....                          | 15 |
| 6.8 Monte Carlo Simulation .....                             | 15 |
| 7. FINDINGS AND BUSINESS INSIGHTS .....                      | 16 |
| 7.1 Which RFM Segment Generates the Most Revenue? .....      | 16 |
| 7.2 Is There Revenue Seasonality? .....                      | 17 |
| 7.3 Which Category Has the Highest Effective Margin? .....   | 17 |
| 7.4 Do Higher Ratings Correlate With Repeat Purchases? ..... | 17 |
| 7.5 Device/Country Conversion .....                          | 18 |
| 7.5.1 Business Implications .....                            | 19 |
| 7.6 Two-Month Revenue Forecast .....                         | 20 |
| 7.6.1 How Management Should Use the Forecast.....            | 20 |
| 7.7 Stockout Risk.....                                       | 21 |
| 7.7.1 Recommended Management Interpretation.....             | 22 |
| 7.8 Impact of Data Quality .....                             | 22 |
| 7.8.1 Data Quality as a Management Control.....              | 22 |
| 8. LIMITATIONS AND THREATS TO VALIDITY .....                 | 22 |
| 8.1 Missing Data.....                                        | 22 |
| 8.2 Duplicate Records.....                                   | 23 |
| 8.3 Returns.....                                             | 23 |
| 8.4 Outliers .....                                           | 23 |
| 8.5 Forecasting Assumptions.....                             | 23 |
| 8.6 Monte Carlo Assumptions .....                            | 23 |
| 9. RECOMMENDATIONS AND NEXT STEPS .....                      | 23 |
| 9.1 Priority 1 Customer Retention .....                      | 23 |

|                                             |    |
|---------------------------------------------|----|
| 9.2 Priority 2 Inventory Optimization ..... | 23 |
| 9.3 Priority 3 Margin Improvement.....      | 24 |
| 9.4 Priority 4 Data Governance .....        | 24 |
| 9.5 Priority 5 Digital Conversion .....     | 24 |
| 10. CONCLUSION .....                        | 24 |
| APPENDIX FINAL ANALYTICAL CHECKLIST .....   | 26 |
| A. Required SQL Deliverables .....          | 26 |
| B. Required Cleaning Deliverables .....     | 26 |
| C. Required NumPy Deliverables .....        | 26 |
| D. Final Repository Structure .....         | 27 |

# **1. EXECUTIVE SUMMARY**

## **1.1 Project Background**

Urban Cart is presented in the supplied report as a fictional online retailer operating across six product categories and fifteen cities worldwide. The organization generates information through customer registrations, orders, products, reviews, and website sessions. The purpose of the project is to transform these operational records into evidence that can support customer management, profitability analysis, demand planning, and data-governance decisions.

The central idea of the project is that raw transactional information should not be treated as automatically reliable. The source document explicitly identifies duplicate records, missing values, inconsistent formats, unusual values, returns, and other quality problems. Consequently, the analytical workflow is designed to connect data engineering and business analysis rather than treating cleaning as a separate technical exercise.

## **1.2 Strategic Questions**

- Who are Urban Cart's most valuable customers, and what patterns are associated with loyalty or churn?
- Which products and categories remain economically attractive after considering cost, discounting, and returns, and can near-term demand be forecast?
- Is the marketing and signup data sufficiently trustworthy for management decisions, and what controls should be introduced when it is not?

## **1.3 Analytical Approach**

The project follows four phases. SQL extracts structured business measures from the SQLite database; Pandas cleans and integrates transactional and external CSV data; NumPy implements RFM segmentation, cosine similarity, regression, and Monte Carlo simulation directly; and the final phase

translates analytical outputs into business insights. The supplied report requires ten SQL queries and eight business questions, with visual evidence supporting the findings.

## **1.4 Main Analytical Evidence Available in the Supplied Report**

The supplied document already contains several visual outputs. The RFM chart on page 13 shows four segments' Champions, Loyal, Potential, and At Risk with Loyal customers contributing the largest displayed net revenue, followed by Champions, Potential, and At Risk. This visual should be interpreted as the project's calculated output rather than as a general e-commerce rule.

The review-rating analysis shown on page 15 reports a Pearson correlation of 0.031 between average customer review rating and repeat purchase. The three displayed rating bands have similar repeat-buyer rates, suggesting that the relationship is weak in this dataset. The report correctly emphasizes that correlation does not establish causation.

The device/country visualization on page 16 shows the highest displayed seven-day session-to-purchase conversion for mobile users in France, followed by tablet users in the USA and mobile users in Germany. These results are useful for identifying combinations for deeper investigation, but they should not be interpreted as causal effects without additional controls.

The regression chart on page 17 displays historical monthly revenue together with a fitted trend and two forecast points. The visual suggests a positive long-term direction despite substantial month-to-month variation. The source document appropriately notes that forecast values are estimates rather than guarantees.

## **1.5 Management Value**

Taken together, the workflow gives management a way to connect customer value, product economics, digital engagement, forecasting, inventory risk, and data quality. The strongest value of the project is therefore not one

isolated chart but the integrated chain from imperfect records to defensible decisions.

## **2. INTRODUCTION AND BUSINESS CONTEXT**

### **2.1 Urban Cart**

Urban Cart is described as an online retailer serving fifteen cities and selling six broad categories: Electronics; Home & Kitchen; Apparel; Sports & Outdoors; Beauty & Health; and Books & Media. The business produces multiple forms of digital data. Customer tables describe who buys; order and order-item tables describe what is purchased; product data describes prices and costs; reviews provide customer feedback; and web sessions provide evidence of digital engagement.

From an analytics perspective, these sources represent different levels of observation. A customer is a person-level entity, an order is a transaction-level entity, an order item is a line-level entity, a review is a product-feedback event, and a web session is an engagement event. Keeping these levels distinct is important because joining them without care can multiply rows and distort totals.

### **2.2 Why Business Context Matters**

A technically correct SQL query can still answer the wrong business question. For example, ranking products only by gross sales could favor products that are frequently returned or heavily discounted. Similarly, counting customers without addressing duplicate CRM records could overstate the customer base. The project therefore frames each analytical method around a management decision.

The business context also explains why the project uses several tools. SQL is appropriate for relational extraction and aggregation; Pandas is suitable for data cleaning and reconciliation; NumPy is appropriate for transparent numerical implementation of the required analytical methods.



## **2.3 Business Question 1: Customer Value and Loyalty**

The first question focuses on customer value and loyalty. The supplied report proposes RFM analysis using recency, frequency, and monetary value. Recency measures how recently a customer purchased, frequency measures the number of purchases, and monetary value measures total customer spending. Together, these dimensions provide a compact behavioral profile.

The segmentation should be interpreted as a prioritization framework. Champions may deserve retention investment because they combine strong recent activity, frequency, and spending. At-risk customers may require reactivation strategies. Potential customers can be developed into more valuable segments. The actual segment definitions should be based on the observed customer distribution, as required by the source assignment.

## **2.4 Business Question 2: Profitability and Demand**

The second question goes beyond revenue. Product performance must consider product cost, discounts, and returns. A high-revenue product can produce a weak effective margin if its costs are high or if many units are returned. The project therefore combines category revenue analysis with return-rate analysis and effective-margin reasoning.

Demand planning is supported by two numerical methods: normal-equation regression for near-term revenue forecasting and Monte Carlo simulation for stockout risk. The methods are deliberately transparent and implemented with NumPy rather than high-level machine-learning shortcuts.

## **2.5 Business Question 3: Data Trustworthiness**

The third question treats data quality as a business issue. The legacy customer export includes multiple date formats, inconsistent capitalization, duplicate or near-duplicate customers, missing emails, test accounts, and blank rows. The external product catalog also uses different naming conventions. These problems can directly affect customer counts, revenue attribution, segmentation, and product reconciliation.

### 3. DATA DESCRIPTION

#### 3.1 Data Sources

The project uses three principal sources: the main ecommerce.db SQLite database, a historical legacy\_customers\_export.csv file, and an external product\_catalog\_2024.csv file. The two CSV sources are intentionally outside the database so that integration and reconciliation become explicit parts of the assignment.

| Source                      | Role in the analysis               |
|-----------------------------|------------------------------------|
| ecommerce.db                | Main transactional SQLite database |
| legacy_customers_export.csv | Historical CRM/customer export     |
| product_catalog_2024.csv    | External product catalog           |

#### 3.2 Database Tables

| Table       | Main purpose                                   |
|-------------|------------------------------------------------|
| customers   | Customer demographic and signup information    |
| products    | Product, category, price, and cost information |
| orders      | Customer order transactions                    |
| order_items | Products and quantities within orders          |

|              |                                 |
|--------------|---------------------------------|
| reviews      | Product ratings and review text |
| web_sessions | Website engagement information  |

The source report identifies several known quality issues in the transactional database. Approximately 3% of customer city values are missing, approximately 6% of customer ages are missing, approximately 1% of product prices contain extreme outliers, approximately 0.8% of order-item records are exact duplicates, negative order-item quantities represent returns, some review ratings are outside the expected range, and approximately 20% of review text values are missing.

These issues have different analytical consequences. Missing city values mainly affect geographic segmentation and payment-preference interpretation. Missing ages affect demographic profiling. Price outliers can distort revenue and margin calculations. Exact duplicates can inflate item quantities and revenue. Negative quantities require special treatment because they represent returns rather than ordinary sales.

Review-quality problems require another distinction: a missing review text does not necessarily mean a missing rating. The source report proposes keeping the rating while flagging missing review text. This preserves usable information without inventing content.

### **3.3 External Customer Data**

- Four different date formats must be standardized.
- Name capitalization is inconsistent.
- The header contains a trailing whitespace issue.
- Approximately 5% of customers are duplicate or near-duplicate records.
- Some email values are missing.
- Test accounts and blank rows are present.

The report explicitly requires these problems to be identified and documented rather than silently corrected. This is important for auditability: a future reviewer should be able to understand which records were transformed, which were excluded, and which were retained with flags.

### **3.4 External Product Catalog**

The external catalog uses different naming conventions from the internal database. In the source document, the internal field `product_id` corresponds conceptually to the external field `SKU`, while `category` corresponds to `dept`. The catalog also contains approximately twelve supplier-only SKUs that do not exist in the Urban Cart database.

The reconciliation task should therefore produce three clear populations: products found in both sources, products found only in the internal database, and products found only in the external catalog. A fourth useful interpretation is that supplier-only products may represent potential future assortment rather than data errors. They should not be inserted into transactional history merely because they appear in the external catalog.

### **3.5 Data Lineage**

A clean analytical workflow should preserve lineage from raw source to processed output. The repository structure supplied in the report includes raw CSV files and processed files such as `clean_customers.csv`, `clean_orders.csv`, `clean_order_items.csv`, and `clean_products.csv`. This separation supports reproducibility and reduces the risk of overwriting original evidence.

The final report should explain transformations in plain language. For example, if a date is standardized, the original raw value should remain available in the raw dataset while the processed field contains a consistent datetime representation.

### **3.6 Data Dictionary Considerations**

For each analytical field, the project should record its meaning, expected type, source, cleaning rule, and downstream use. This is especially important for fields such as quantity because positive values represent sales while

negative values represent returns. Similarly, a rating outside the expected range should be flagged rather than automatically interpreted as a valid score.

## **4. SQL EXTRACTION AND METHODOLOGY**

The source assignment requires ten analytical SQL queries. Each query must include a one-line comment explaining its purpose. The queries should be reproducible against the SQLite database and should avoid unnecessary duplication caused by joins.

### **4.1 Revenue by Category**

The first query calculates total revenue, order count, and average order value for each product category after discounts. Its management purpose is to identify which categories contribute most to sales. The calculation should use the correct grain: revenue should be based on order-item values rather than multiplying order-level totals by product rows.

### **4.2 Top 20 Customers**

The second query identifies the twenty customers with the highest lifetime spending. It joins customers to orders and then to order\_items. Customer city and signup date are included so that the output can support both value analysis and demographic interpretation.

### **4.3 Monthly Revenue Trend**

The third query aggregates revenue by month for the most recent twenty-four months. A window function such as LAG() compares each month with the previous month. This supports a clearer interpretation of growth and decline than a single total would provide.

### **4.4 Category Return Rate**

Return rate is calculated using order items with negative quantities. The query should make the definition explicit because return rate can be expressed in several ways. The source assignment specifically directs attention to negative quantities, so the report should preserve that interpretation.

## **4.5 Consistent Customers**

The fifth query identifies customers who purchased in each of the most recent three calendar quarters. These customers provide evidence of sustained purchasing behavior. The result can later be compared with RFM segments to see whether consistently active customers are concentrated in higher-value groups.

## **4.6 Highly Rated Products**

The sixth query returns the ten products with the highest average review rating while requiring at least fifteen reviews. The minimum review threshold is important because a product with one perfect rating should not automatically outrank a product with many strong ratings.

## **4.7 Device Engagement**

The seventh query compares average session duration and pages viewed by device type for customers who have made at least one purchase. This creates a bridge between web engagement and customer activity.

## **4.8 Product Ranking Within Category**

Products are ranked by revenue within their respective categories using `RANK()` or `DENSE_RANK()`. Category-level ranking is more informative than a single global ranking because it identifies leading products within each competitive group.

## **4.9 Payment Method Mix**

Payment methods are compared by country to identify geographic differences in payment preferences. The output can inform checkout optimization and localization decisions, but the report should distinguish descriptive patterns from causal explanations.

## **4.10 Additional Leadership Analysis**

The tenth query is reserved for an additional management question selected because it has direct business value. A strong choice should complement not duplicate the other nine analyses. For example, leadership

could investigate the relationship between discounting and return behavior, provided the required fields support that analysis.

## **4.11 SQL Quality Controls**

- Check that joins preserve the intended observation grain.
- Use explicit aliases for calculated fields.
- Handle nulls intentionally rather than allowing them to disappear silently.
- Document whether revenue is gross, discounted, or net of returns.
- Validate totals against independent calculations where possible.

## **5. DATA CLEANING AND INTEGRATION**

The supplied report identifies data cleaning as one of the largest sections because the assignment requires five to six pages, before/after evidence, and explanations of what was not fixed. Cleaning should therefore be treated as a documented analytical process rather than a series of hidden edits.

### **5.1 Date Cleaning**

The legacy customer file contains four date formats. The objective is to convert these values into a single standardized datetime field suitable for time-series analysis. A robust process first identifies the recognized patterns, parses them consistently, and records invalid or ambiguous values rather than forcing an arbitrary interpretation.

Date standardization matters because customer signup dates may be used in demographic analysis, cohort-style comparisons, and reconciliation. If the same calendar date is represented in multiple formats, direct string comparisons can incorrectly suggest that two records are different.

### **5.2 Before/After Evidence**

The final report should show representative examples of raw date strings and their standardized equivalents. The evidence does not need to list every row; a small sample plus counts of successfully parsed and unresolved records provides a transparent audit trail.

## 5.3 Duplicate Customers

Duplicate detection is performed in two stages. Exact duplicates contain identical identifying information and can often be removed after confirming that they do not represent legitimate repeated events. Near duplicates require more care because small differences in capitalization, whitespace, spelling, or formatting may hide the fact that two records represent the same customer.

The source report recommends using email, normalized name, signup date, and other available customer attributes. Name capitalization should not cause the same person to be treated as multiple customers. At the same time, name similarity alone should not be used as proof of identity when common names are involved.

## 5.4 Normalization Principles

- Trim leading and trailing whitespace.
- Standardize capitalization for comparison fields.
- Normalize email values where appropriate.
- Create comparison keys without destroying the original raw fields.
- Flag uncertain matches for review rather than automatically merging them.

The approximately 5% duplicate or near-duplicate rate reported for the legacy customer file is therefore a material governance issue. It can affect customer counts, frequency measures, and lifetime spending if multiple records are allowed to represent the same individual.

## 5.5 Missing Values

Missing values should be handled according to the semantic meaning of each variable. The source report gives age as an example that may be imputed or flagged, while city may be retained as missing or represented as unknown. Gender should be retained as unknown where it is not available. Review text can remain missing while the rating is preserved.

The assignment explicitly prohibits blindly replacing missing values with zero. This rule is important because zero often has a real business meaning. For



example, an age of zero is not a reasonable substitute for an unknown age, and a missing city should not be interpreted as a city with no purchases.

## **5.6 Outliers**

Approximately 1% of product prices are described as extreme outliers. Outlier handling should begin with investigation rather than deletion. The analyst should determine whether an extreme price is a legitimate premium product, a unit mismatch, or a data-entry problem. Only after that interpretation should a record be corrected, excluded, or flagged.

## **5.7 Returns**

Negative order-item quantities represent returns. Returns should not simply be converted to positive quantities because doing so would reverse their economic meaning. Instead, they should be incorporated into net-revenue and return-rate calculations according to the business definition used by the project.

## **5.8 Product Reconciliation Workflow**

The external catalog and internal products table use different field names and potentially different naming conventions. A reconciliation workflow should first standardize identifiers and text fields, then match records using the strongest available key. The output should explicitly separate matched products, internal-only products, and external-only products.

The approximately twelve supplier-only SKUs should be reported rather than inserted into historical transactions. Their presence may be commercially useful, but they do not prove that Urban Cart sold those products.

## **5.9 Integration Validation**

- Compare record counts before and after integration.
- Check that product identifiers remain unique where expected.
- Verify that joins do not multiply order-item rows.
- Compare revenue totals before and after cleaning.
- Document every exclusion and transformation.
- Keep raw files unchanged so results can be reproduced.

## **5.10 Processed Outputs**

The required repository structure includes processed customer, order, order-item, and product files. These outputs create a stable layer for downstream SQL/Pandas/NumPy analysis and make it easier for another group member or lecturer to reproduce the work.

## **5.11 Why Cleaning Can Change a Conclusion**

The assignment requires at least one example where cleaning materially changes a business conclusion. The supplied report already includes a visual on page 17 comparing raw and post-deduplication net revenue by category. The bars show that exact-order-item deduplication produces small but visible reductions in reported net revenue.

This example illustrates an important principle: even a relatively small duplicate rate can change a management metric. If a category is close to another category in performance, a duplicated set of order items could alter its ranking. The correct conclusion is therefore not that every difference is large, but that data-quality controls are necessary before making close comparisons.

## **5.12 What Was Not Fixed**

A transparent report should identify unresolved limitations. For example, uncertain near-duplicate customers may be flagged rather than merged when the available evidence is insufficient. Missing demographic values may remain missing. Supplier-only SKUs may remain outside the transactional dataset. Such decisions protect the analytical integrity of the project.

## **5.13 Reproducibility**

All cleaning steps should be encoded in the notebook or scripts rather than performed manually. The source repository includes analysis.ipynb and processed CSV outputs, supporting a reproducible workflow.

# **6. NUMPY ANALYTICAL METHODS**

## **6.1 RFM Segmentation**

RFM measures customer value through Recency, Frequency, and Monetary value. The source assignment defines recency as the difference between the current date and the customer's last purchase date, frequency as the number of purchases, and monetary value as total customer spending.

Each dimension is converted into a score using NumPy arrays. The combined score can be expressed as  $\text{RFMScore} = \text{RScore} + \text{FScore} + \text{MScore}$ . The final segmentation should be determined from the actual distribution of customers rather than from arbitrary universal thresholds.

## **6.2 Interpreting RFM**

RFM is valuable because it distinguishes customers who may have similar total spending but different current behavior. A customer with high spending but no recent activity may need a different strategy from a customer with moderate spending who purchases consistently.

The supplied RFM visualization on page 13 displays four segments. Loyal customers have the largest displayed net revenue, followed by Champions, Potential, and At Risk. This provides a useful management starting point: retention activity should protect the value already concentrated in the strongest segments while targeted reactivation can address the at-risk population.

## **6.3 Customer-Product Interaction Matrix**

The recommendation component represents customer-product interactions numerically. Each row can represent a customer and each column a product, with values representing the interaction measure selected by the assignment. This representation makes it possible to compare customers or products mathematically.

## **6.4 Cosine Similarity**

The source report requires direct NumPy computation of cosine similarity rather than sklearn. The formula is the dot product of two vectors divided by the product of their norms. In practical terms, the method compares the direction of interaction patterns rather than their absolute size.

## **6.5 Recommendation Output**

The assignment requires three recommended products for five sample customers. Recommendations should be generated from similarity patterns and

should avoid recommending products that the customer has already purchased when the business objective is discovery.

## **6.6 Business Interpretation**

A recommendation system can support cross-selling and personalized merchandising. However, recommendations are only as useful as the interaction data. Sparse purchase histories, duplicate customers, and inconsistent product identifiers can weaken similarity calculations. This is another example of why data cleaning and modeling cannot be separated.

## **6.7 Normal-Equation Regression**

The regression model uses the normal equation  $\beta = (X^T X)^{-1} X^T y$  and NumPy linear-algebra operations. The coefficient of determination is calculated manually using  $R^2 = 1 - SS_{\text{res}}/SS_{\text{tot}}$ . The model is then used to forecast revenue for the following two months.

The supplied regression visualization on page 17 shows a positive fitted trend against highly variable historical monthly revenue. The chart supports the interpretation that the model captures an overall upward direction, but the visible volatility demonstrates why the forecast should be accompanied by uncertainty rather than presented as a guaranteed value.

## **6.8 Monte Carlo Simulation**

The assignment requires at least 5,000 simulation trials to estimate stockout probabilities for selected products. The output should include a simulated demand distribution, stockout probability, confidence interval, and recommended reorder point.

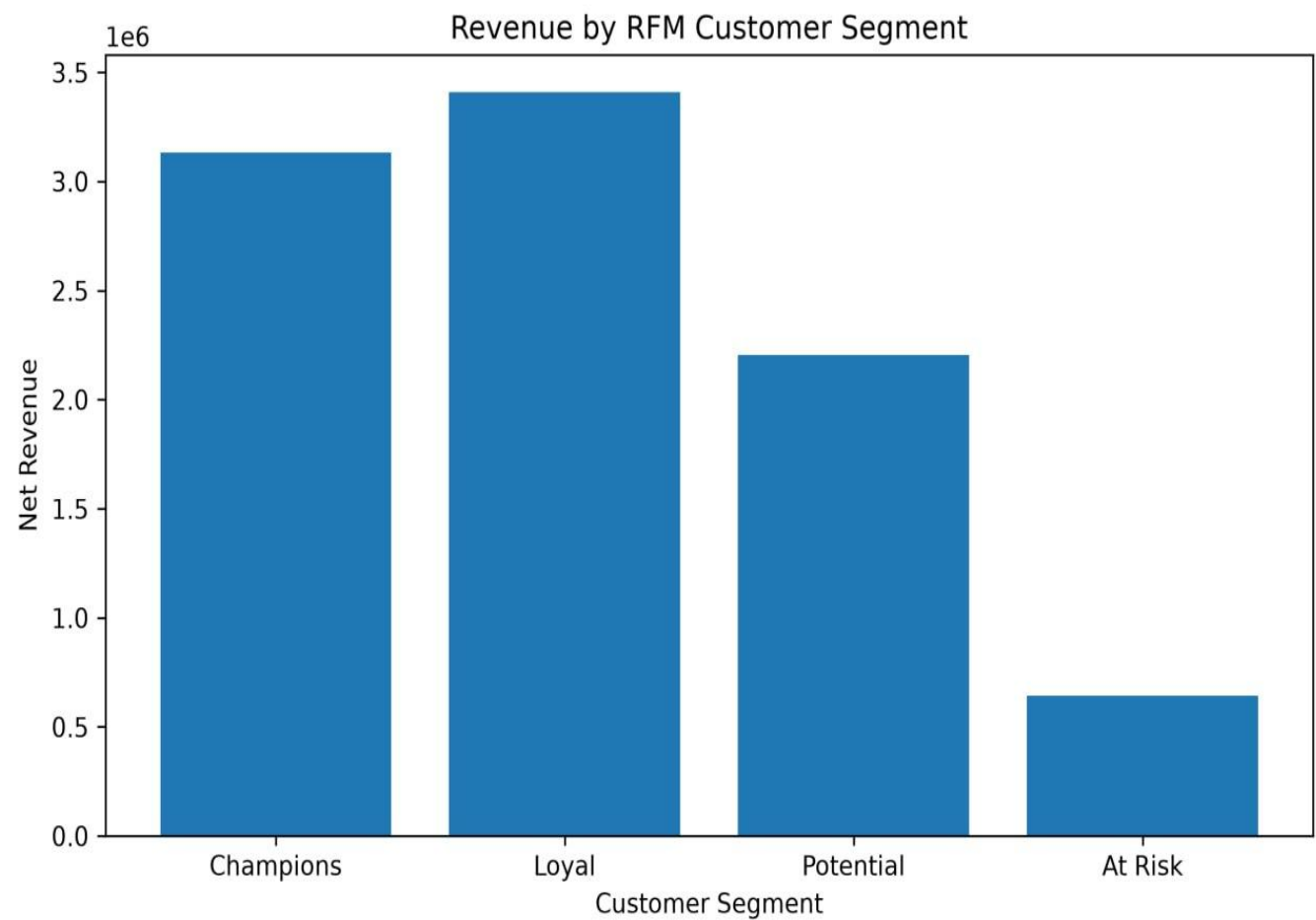
The simulation is especially useful when demand is uncertain. Instead of asking only what the average demand might be, management can ask how frequently simulated demand exceeds available stock. This supports risk-based inventory decisions.

# 7. FINDINGS AND BUSINESS INSIGHTS

## 7.1 Which RFM Segment Generates the Most Revenue?

The RFM chart supplied in the original report shows Loyal customers as the largest revenue-contributing segment, with Champions next, followed by Potential and At Risk. This indicates that revenue is not evenly distributed across behavioral groups. The result supports targeted retention because the high-value segments account for a substantial share of displayed net revenue.

The analysis should be extended by comparing age, gender, city, and purchasing frequency across the segments. These comparisons can reveal whether the strongest segments are concentrated in particular demographic or geographic groups. The source report does not provide the underlying numerical demographic tables, so those details should be populated from the actual cleaned dataset rather than invented.



## **7.2 Is There Revenue Seasonality?**

The monthly revenue analysis is designed to identify recurring increases, decreases, seasonal peaks, and seasonal declines. The source document specifies a rolling average or decomposition approach. The correct interpretation should distinguish a genuine repeating seasonal pattern from a one-time spike.

A useful management output is a monthly chart with a trend line or rolling average. If peaks recur in similar months, inventory and marketing plans can be adjusted. If peaks occur only once, management should investigate the underlying event before labeling the pattern as seasonality.

## **7.3 Which Category Has the Highest Effective Margin?**

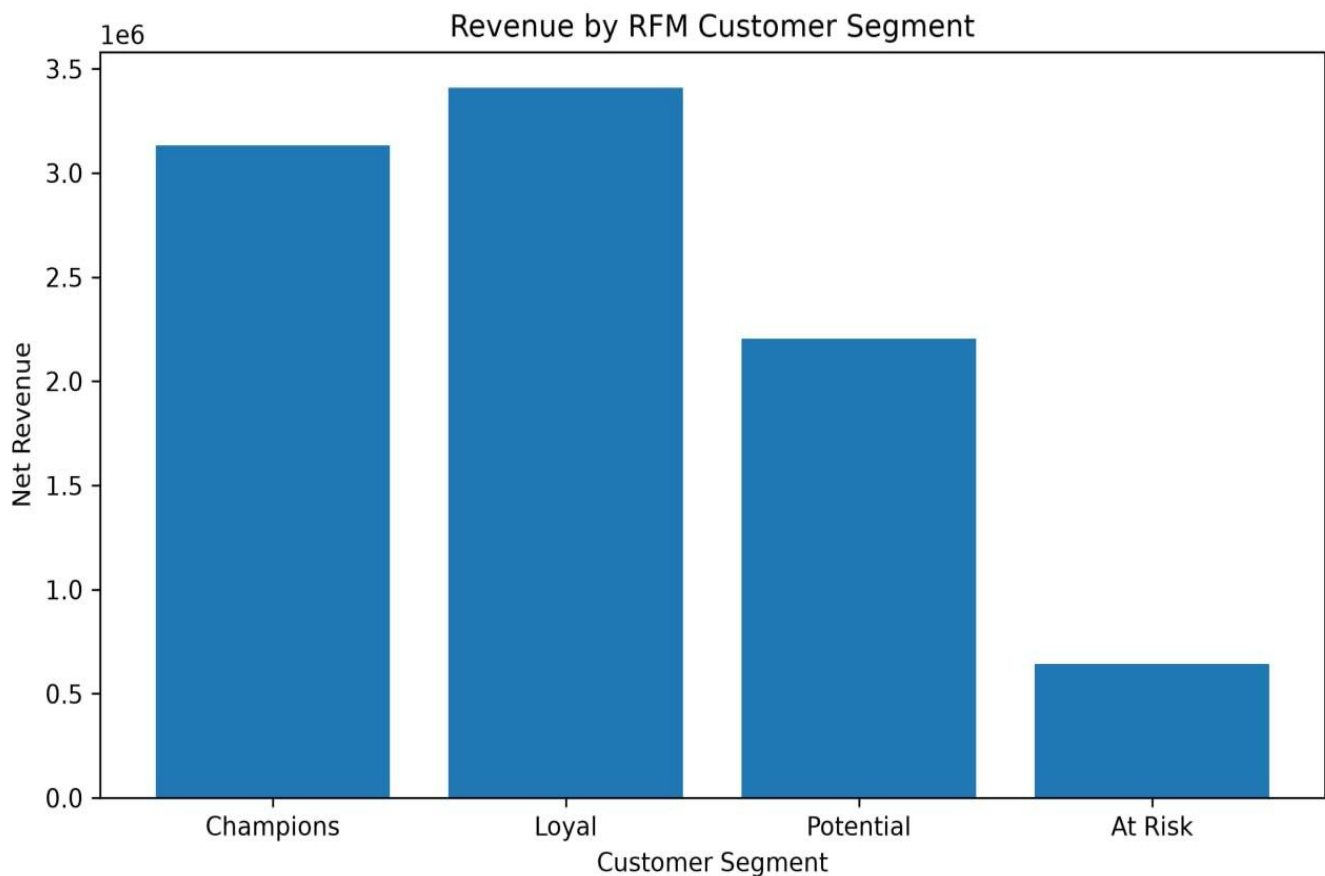
The project defines effective margin using net revenue and net cost, while accounting conceptually for product cost, discounts, and returns. This is more decision-relevant than revenue alone because it identifies categories that retain economic value after the main revenue-reducing factors are considered.

The final report should identify the highest-margin category from the actual calculated output and explain the business drivers. The supplied source does not state the final category name, so this expanded report intentionally does not invent one.

## **7.4 Do Higher Ratings Correlate With Repeat Purchases?**

The supplied chart reports Pearson  $r = 0.031$ . The displayed repeat-purchase rates across rating bands are close to one another, indicating a very weak linear relationship in this dataset. This means higher ratings alone do not appear to be a strong predictor of repeat purchase behavior.

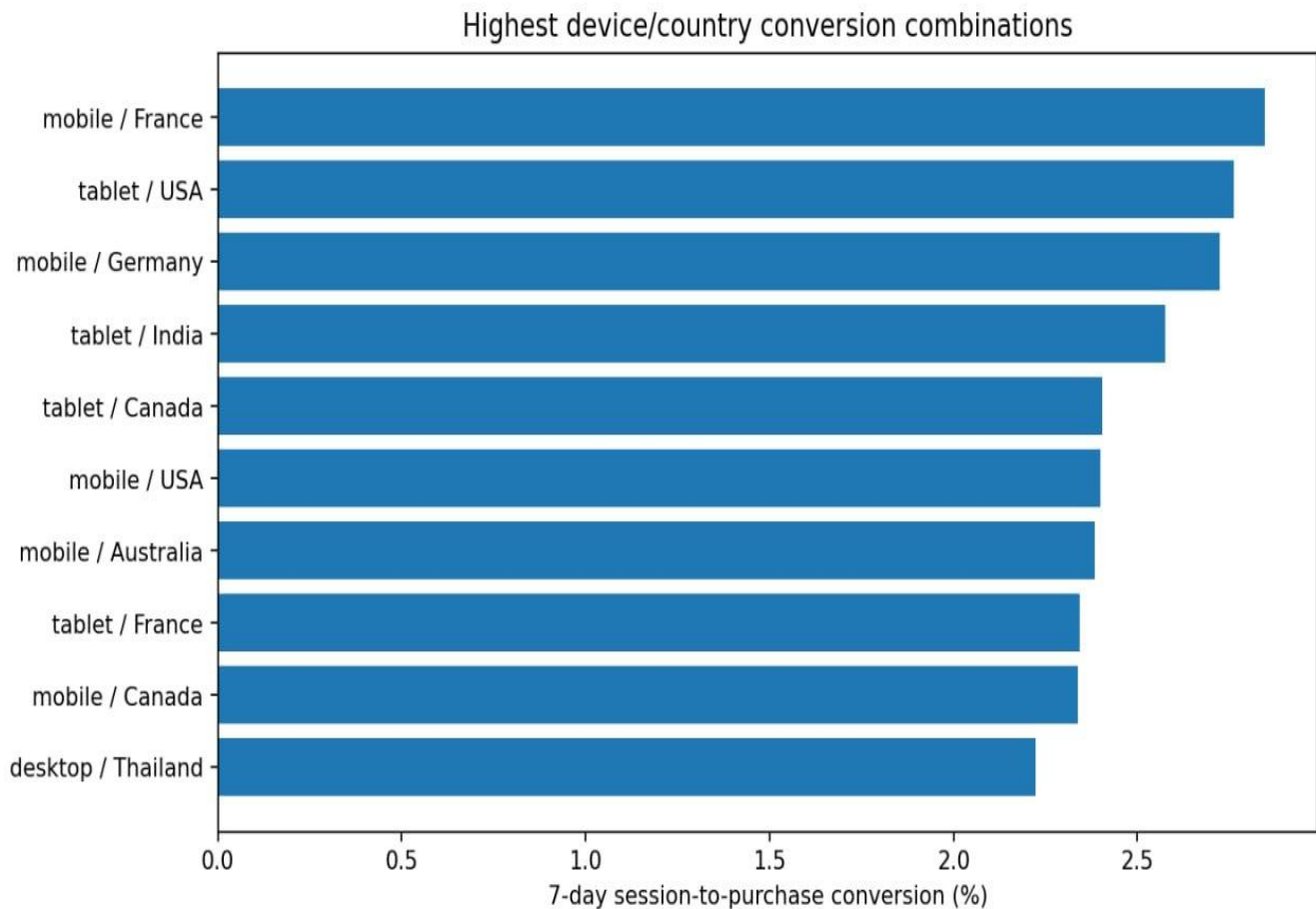
The result should still be interpreted carefully. A weak correlation does not prove that reviews have no business value. Reviews may influence conversion, product discovery, or trust in ways not captured by the specific repeat-purchase measure. The source report correctly emphasizes that correlation does not automatically establish causation.



## 7.5 Device/Country Conversion

The supplied visualization compares seven-day session-to-purchase conversion across device and country combinations. The highest displayed combination is mobile / France, followed by tablet / USA and mobile / Germany. Other high-performing combinations shown include tablet / India, tablet / Canada, mobile / USA, mobile / Australia, tablet / France, mobile / Canada, and desktop / Thailand.

These results suggest that conversion performance may vary substantially by the interaction of device and geography. A country-level average alone could hide important device differences. For example, a mobile experience may perform differently from a desktop experience within the same market.



### 7.5.1 Business Implications

- Investigate checkout usability for high-engagement but lower-conversion combinations.
- Compare payment methods and shipping options across countries.
- Review mobile page performance and friction points.
- Use controlled experiments before attributing conversion differences to a single cause.
- Monitor the same combinations over time to distinguish stable patterns from temporary changes.

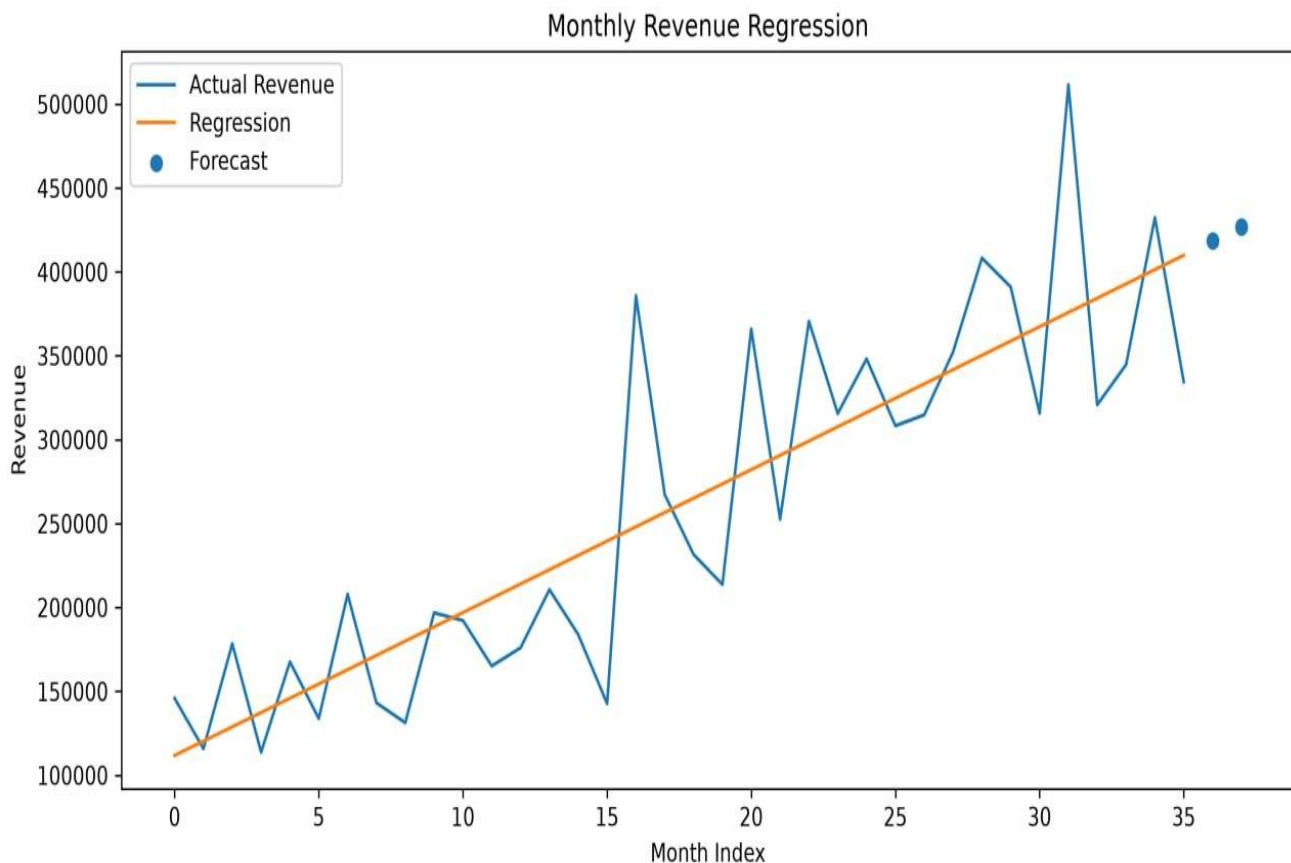
The chart should therefore be used as a prioritization tool. It identifies where management should investigate rather than proving why conversion differs.



## 7.6 Two-Month Revenue Forecast

The regression model is designed to generate revenue projections for the next two months. The supplied chart shows two forecast points beyond the historical series, positioned around the continuing upward trend.

Forecast uncertainty is important because a model summarizes historical patterns rather than controlling future outcomes. Changes in promotions, product availability, customer behavior, returns, seasonality, or external conditions can cause actual revenue to differ from the fitted trend.



### 7.6.1 How Management Should Use the Forecast

- Use the forecast as a planning baseline rather than a guaranteed target.
- Combine the forecast with inventory information before placing large purchase orders.
- Compare actual monthly revenue with forecast values and record forecast error.

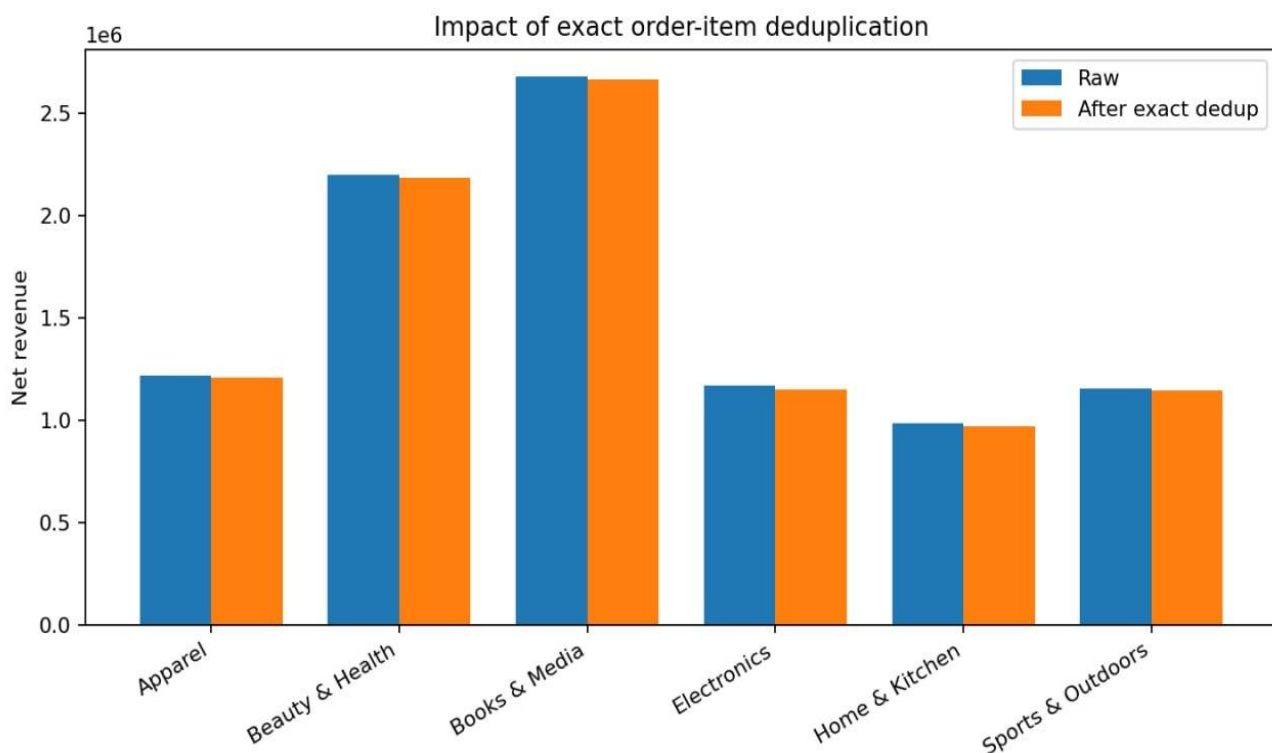
- Re-estimate the model when new data becomes available.
- Investigate major deviations instead of simply treating them as model failure.

A practical dashboard would show actual revenue, fitted revenue, forecast revenue, and an uncertainty range. This makes the difference between observed and estimated values visually clear.

## 7.7 Stockout Risk

The Monte Carlo simulation is required to identify the five products with the highest stockout risk. The simulation should use at least 5,000 trials and estimate the probability that uncertain demand exceeds available stock.

A stockout probability is more actionable when paired with a reorder point. The reorder point should reflect expected demand during the relevant replenishment period plus an allowance for uncertainty. Because the source document does not provide the underlying product-level simulation table, the final five product names should be inserted directly from the completed notebook rather than guessed.



### **7.7.1 Recommended Management Interpretation**

- High stockout probability means the product deserves inventory attention.
- A high probability with low revenue may still be less urgent than a moderate probability on a strategically important product.
- Reorder points should be reviewed when demand volatility changes.
- Simulation assumptions should be documented because different demand distributions produce different risk estimates.
- The strongest risks should be reviewed together with supplier lead time and current inventory.

### **7.8 Impact of Data Quality**

The assignment requires at least one example in which cleaning materially changes a business conclusion. The supplied report provides evidence through a raw-versus-deduplicated revenue comparison. The post-deduplication bars are slightly lower than the raw values across the displayed categories.

The most important lesson is methodological: the magnitude of a correction should not be judged only by its percentage. If two categories have very similar performance, a small correction can change their relative order. Conversely, a small correction may have little effect on a broad strategic conclusion. Both possibilities should be checked.

#### **7.8.1 Data Quality as a Management Control**

Data quality should be monitored continuously rather than only when a report is prepared. Useful controls include duplicate-rate monitoring, missing-value dashboards, range checks for prices and ratings, validation of foreign-key relationships, and reconciliation between transactional totals and management reports.

This approach moves data quality from a one-time cleaning activity toward data governance. It also reduces the risk that future decisions will be based on a dataset that has silently degraded.

## **8. LIMITATIONS AND THREATS TO VALIDITY**

### **8.1 Missing Data**

Missing demographic values may affect customer segmentation. If missingness is concentrated in particular groups, simple summaries may not represent the full population.

## **8.2 Duplicate Records**

Duplicate customers can distort customer-level spending and purchase frequency. Exact duplicates are easier to detect than near duplicates, which require judgment and may remain flagged rather than merged.

## **8.3 Returns**

Negative quantities require careful treatment because gross sales and realized revenue are not necessarily the same. Any margin or revenue metric should state how returns are handled.

## **8.4 Outliers**

Extreme product prices can disproportionately affect revenue and margin calculations. The analyst should investigate whether an outlier is legitimate before removing it.

## **8.5 Forecasting Assumptions**

Regression assumes historical information is useful for future revenue. Structural changes or strong seasonality may weaken that assumption.

## **8.6 Monte Carlo Assumptions**

Stockout probabilities depend on the assumed demand distribution and uncertainty. Simulation output should therefore be presented together with its assumptions.

# **9. RECOMMENDATIONS AND NEXT STEPS**

The supplied report proposes three to five concrete priorities. The recommendations below preserve that structure while expanding the implementation logic.

## **9.1 Priority 1 Customer Retention**

Target high-value RFM customers with personalized retention strategies. Champions and Loyal customers should receive protection-oriented treatment, while At Risk customers can receive carefully targeted reactivation offers.

## **9.2 Priority 2 Inventory Optimization**

Use stockout probabilities and reorder points to prioritize replenishment. Inventory decisions should combine simulated demand risk with commercial importance and supplier lead time.

### **9.3 Priority 3 Margin Improvement**

Investigate products that generate strong revenue but weak effective margins. Discounting, cost structure, and return behavior should be examined together.

### **9.4 Priority 4 Data Governance**

Create standardized customer and product data-management rules. Duplicate detection, missing-value monitoring, date standards, and identifier reconciliation should become routine controls.

### **9.5 Priority 5 Digital Conversion**

Investigate high-engagement and low-conversion device/country combinations. Focus on checkout friction, payment methods, shipping expectations, and device usability before making causal claims.

## **10. CONCLUSION**

The Urban Cart project demonstrates a complete analytics workflow beginning with imperfect raw data and ending with business recommendations. The project integrates SQL, Pandas, and NumPy rather than treating them as independent tools. SQL provides structured extraction from the transactional database, Pandas manages cleaning and integration, and NumPy provides direct numerical implementations of segmentation, similarity, regression, and simulation.

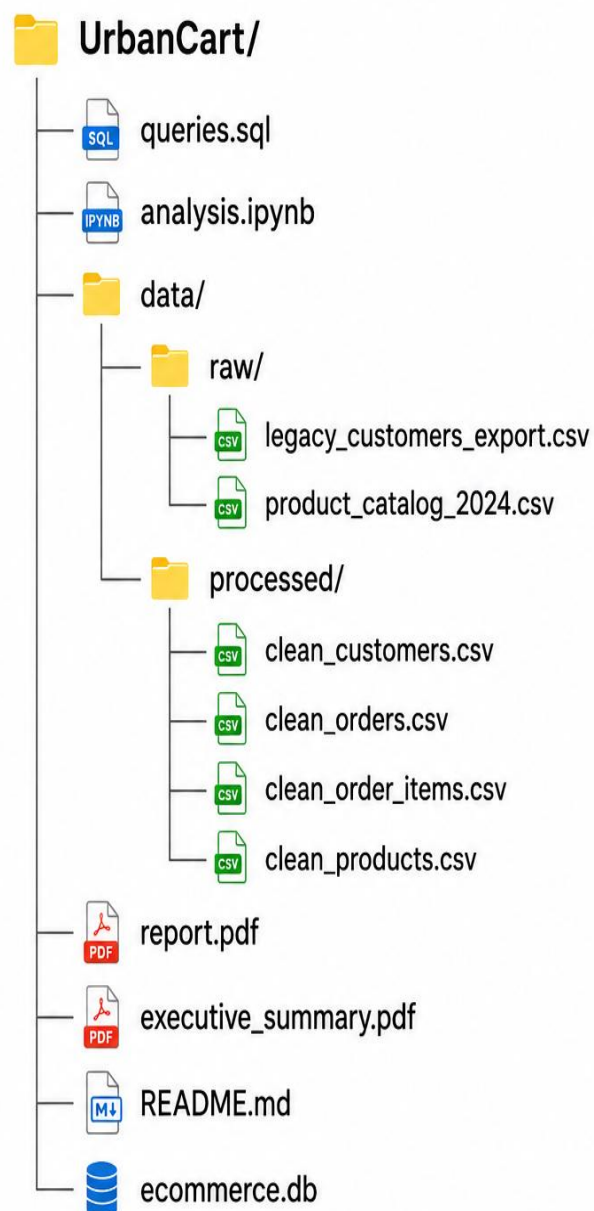
The strongest methodological lesson is that data quality is part of business analysis. Duplicate customers, missing information, inconsistent formats, product-price outliers, and returns can materially change analytical conclusions. Reliable decision-making therefore requires both analytical methods and disciplined preparation of the data.

The supplied visual evidence provides several useful findings. Loyal customers represent the largest displayed RFM revenue segment. The review-rating relationship with repeat purchase is very weak, with Pearson  $r = 0.031$ . Mobile/France is the highest displayed device/country conversion combination.

Historical revenue shows an overall upward fitted trend, although substantial variation remains.

The project should finish by connecting these results to the actual calculated outputs in the notebook. Where the source document leaves a result as '[Calculated from database]', the final submission should insert the verified value from the completed dataset rather than relying on assumptions.

Overall, the Urban Cart workflow provides a strong foundation for evidence-based decisions about customer retention, inventory, profitability, digital conversion, and data governance.



# APPENDIX FINAL ANALYTICAL CHECKLIST

## A. Required SQL Deliverables

- Revenue by category
- Top 20 customers
- Monthly revenue trend with LAG() or equivalent
- Category return rate
- Consistent customers across three quarters
- Top ten highly rated products with at least 15 reviews
- Device engagement
- Product ranking within category
- Payment method mix by country
- One additional leadership question

## B. Required Cleaning Deliverables

- Standardize four legacy date formats
- Detect exact and near-duplicate customers
- Handle missing values according to variable meaning
- Investigate price outliers
- Treat negative quantities as returns
- Reconcile internal products with the external catalog
- Provide before/after evidence
- Document what was not fixed

## C. Required NumPy Deliverables

- RFM segmentation
- Cosine-similarity recommendation system
- Normal-equation regression and  $R^2$
- At least 5,000 Monte Carlo trials
- Three products for Phase 3 simulation
- Five strongest stockout risks for Phase 4

## **D. Final Repository Structure**

The supplied report specifies a repository containing queries.sql, analysis.ipynb, raw CSV files, processed CSV files, report.pdf, executive\_summary.pdf, README.md, and ecommerce.db. The completed submission should keep these files clearly organized and reproducible.